

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 2

Mock Interview

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO5: Students will be able to evaluate advanced machine learning concepts such as reinforcement learning, Monte Carlo methods, sampling techniques, policy iteration, value iteration, and temporal difference learning. They will understand computational learning theory concepts including VC dimension, sample complexity, and mistake-bound analysis. Students will be capable of selecting and optimizing advanced learning strategies for intelligent systems and complex applications. (BL-6)

Assessment Tool Weightage: 20%

QUESTIONS:

Q1. Autoencoder Design

- What is an autoencoder and what are its main components in a neural network architecture?
- How would you design an autoencoder for reducing the dimensionality of a high-dimensional dataset?
- What factors would you consider when deciding the size of the encoder and decoder layers?
- Why is the bottleneck layer important in an autoencoder design?
- How would you choose an appropriate activation function for the encoder and decoder of an autoencoder?

Q2. Stochastic Encoders and Decoders

- What is a stochastic encoder, and how does it differ from a deterministic encoder?
- Why is randomness introduced into the encoding process of a stochastic autoencoder?
- How does a stochastic decoder reconstruct data from a probabilistic latent representation?
- What are the advantages of using stochastic encoders and decoders in generative models?
- How does a stochastic encoder represent uncertainty in the input data?

Q3. Deep Belief Network vs Deep Boltzmann Machine

- What is the main architectural difference between a Deep Belief Network (DBN) and a Deep Boltzmann Machine (DBM)?
- How are connections between hidden layers organized differently in a DBN and a DBM?
- Why is a DBN considered a hybrid model containing both directed and undirected connections?
- How does a DBM differ from a DBN in terms of the direction of connections between layers?
- How does the training procedure of a DBN differ from that of a DBM?

Q4. Generative Adversarial Networks

- What is a Generative Adversarial Network, and what are its two main components?
- How does the generator in a GAN create new data samples?

- What is the primary role of the discriminator in a GAN?
- How do the generator and discriminator compete with each other during GAN training?
- Why is balancing the generator and discriminator important for stable GAN training?

Q5. Integrated Deep Learning Solution

- What do you understand by an integrated deep learning solution, and what are its major components?
- How would you combine different deep learning models to solve a complex real-world problem?
- How would you decide whether to use an autoencoder, DBN, DBM, or GAN in an integrated solution?
- What factors should be considered when designing an end-to-end deep learning pipeline?
- How can feature extraction and representation learning be integrated into a deep learning solution?

Code of Conduct:

I Haresh Kumar N L (192425009) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Answers:

Q1. Autoencoder Design

1. What is an autoencoder and what are its main components?

An autoencoder is a neural network that learns to compress input data and reconstruct it. Its main components are the encoder, bottleneck, and decoder.

2. How would you design an autoencoder for dimensionality reduction?

I would use an encoder to reduce the input dimensions into a smaller latent representation, and a decoder to reconstruct the original data.

3. What factors would you consider when deciding the size of encoder and decoder layers?

I would consider the input size, complexity of the data, desired compression level, and reconstruction accuracy.

4. Why is the bottleneck layer important?

The bottleneck layer contains the most important features of the input data in a low-dimensional representation.

5. How would you choose activation functions?

I would commonly use ReLU for hidden layers and choose the output activation based on the type of input data, such as sigmoid for normalized data.

Q2. Stochastic Encoders and Decoders

1. What is a stochastic encoder, and how does it differ from a deterministic encoder?

A stochastic encoder produces a probability distribution for the latent representation, while a deterministic encoder produces a fixed latent value.

2. Why is randomness introduced into the encoding process?

Randomness is introduced to represent uncertainty and allow the model to generate different latent representations.

3. How does a stochastic decoder reconstruct data?

The decoder samples from the probabilistic latent representation and uses the sample to reconstruct the input data.

4. What are the advantages of stochastic encoders and decoders?

They help model uncertainty, generate diverse outputs, and improve generative modeling.

5. How does a stochastic encoder represent uncertainty?

It represents the input as a probability distribution, usually using parameters such as mean and variance.

Q3. Deep Belief Network vs Deep Boltzmann Machine — Short Answers

1. What is the main architectural difference between DBN and DBM?

A DBN has both directed and undirected connections, while a DBM has only undirected connections.

2. How are connections between hidden layers organized?

In a DBN, higher hidden layers have directed connections. In a DBM, hidden layers are connected bidirectionally (undirected).

3. Why is a DBN considered a hybrid model?

A DBN is hybrid because it contains both directed and undirected connections.

4. How does a DBM differ from a DBN in connection direction?

A DBM uses undirected connections, allowing information to flow in both directions between layers.

5. How does the training procedure differ?

A DBN is generally trained layer-by-layer, while a DBM is trained jointly as a probabilistic model.

Q4. Generative Adversarial Networks — Short Answers

1. What is a GAN and what are its two main components?

A GAN is a generative model with two networks: the Generator and Discriminator.

2. How does the generator create new data samples?

The generator takes random noise as input and produces new data samples.

3. What is the role of the discriminator?

The discriminator distinguishes between real and generated data.

4. How do the generator and discriminator compete?

The generator tries to fool the discriminator, while the discriminator tries to identify fake data.

5. Why is balancing them important?

A proper balance helps achieve stable training and realistic generated samples.

Q5. Integrated Deep Learning Solution — Short Answers

1. What is an integrated deep learning solution?

It combines different deep learning models and techniques to solve a complex problem.

2. How would you combine different models?

I would use different models in different stages, such as feature extraction, classification, and generation.

3. How would you choose between Autoencoder, DBN, DBM, and GAN?

Use an Autoencoder for compression, DBN/DBM for representation learning, and GAN for data generation.

4. What factors should be considered in an end-to-end pipeline?

Consider data quality, preprocessing, model selection, accuracy, computational cost, and evaluation.

5. How can feature extraction and representation learning be integrated?

A deep learning model can automatically learn useful features, which can then be passed to another model for the final task.